LLM-Informed Bayesian Priors for Reducing Experimental Cost in Behavioral Economics

Bo Feng
bfeng66@gatech.edu
Georgia Institute of Technology
CSE 8803 IUQ — Spring 2026

March 2, 2026

Abstract

Human experiments in behavioral economics are expensive, yet large language models (LLMs) exhibit qualitatively human-like behavior in classic economic games [Horton, 2023] and encode knowledge equivalent to dozens of human observations for economic variables [Gao et al., 2026]. We propose a Bayesian framework in which LLM simulations serve as an informative prior and real human observations provide the likelihood, keeping the two data sources in strictly separate roles to avoid methodological circularity. Our primary metric is the *sample size reduction ratio* $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}}$: the minimum human observations needed to reach a target posterior precision under the LLM prior versus a flat prior. We implement a flat-prior MCMC baseline and an LLM-informed SVGD frontier method, and evaluate both on dictator-game and ultimatum-game data from Horton [2023].

1 Review of the Literature

Horton [2023]. Horton introduces "homo silicus": LLMs queried as simulated experimental subjects replicate qualitative findings of dictator and ultimatum games, yielding sharing-ratio distributions broadly consistent with published human data. The analysis is purely qualitative with no uncertainty bounds, but it establishes the elicitation protocol and the human-response dataset we use as ground truth for evaluation.

Gao et al. [2026]. Gao, Han, and Liang define the *equivalent sample size* (ESS) of an LLM's pretrained knowledge as the task-specific human observations needed to match LLM prediction error, finding values of dozens to hundreds for certain economic variables. ESS is a frequentist prediction metric with no direct Bayesian interpretation; our work operationalizes the ESS intuition inside a Bayesian pipeline, translating it into the sample-size reduction ratio ρ.

Ludwig et al. [2025]. Ludwig, Mullainathan, and Rambachan show that valid LLM-assisted inference requires strict separation of LLM training data from the researcher's sample ("no training leakage") and a held-out validation sample for estimation. Although their framework targets prediction/estimation rather than Bayesian priors, our prior/likelihood role separation directly implements the no-leakage requirement, and a held-out subset of the Horton data serves as the validation sample.

Shorinwa et al. [2025]. Shorinwa et al. provide a comprehensive taxonomy of UQ methods for LLMs—covering epistemic/aleatoric decomposition, calibration metrics, and conformal prediction—though the focus is uncertainty *of the LLM* rather than of downstream parameters θ. Their calibration framework supplies our evaluation metric (empirical coverage of 95% credible intervals) and their epistemic/aleatoric distinction motivates the failure-mode sensitivity analysis.

Gui and Toubia [2025]. Gui and Toubia show that standard blind-prompting in LLM experiments violates unconfoundedness—treatment variations alter unspecified contextual variables, biasing simulated responses—with no general solution when unblinding itself changes behavior. This motivates our prior-misspecification sensitivity analysis: a biased LLM prior proxies for blind-prompting confounding, and we measure the resulting degradation in posterior coverage.

LLM-elicited priors: closest related work. Three concurrent works use LLMs to construct Bayesian priors; we address each in turn.

Capstick et al. [2025] propose AutoElicit, which queries an LLM for the mean and standard deviation of a Gaussian prior over each feature of a *linear predictive model*, then forms a K=100-component Gaussian mixture prior and runs MCMC inference. The evaluation metric is posterior predictive accuracy on clinical datasets (UTI, heart disease). Our approach differs along four dimensions. (i) *Domain and generative model.* AutoElicit targets regression/classification weights $\theta \in \mathbb{R}^d$ under a linear likelihood; we target behavioral parameters $\theta = (\alpha, \beta)$ of a Beta likelihood for sharing ratios in economic games—a bounded, compositional parameter with no natural Gaussian form. (ii) *Elicitation mechanism.* AutoElicit asks the LLM to report (μ_k, σ_k) directly; we instead ask the LLM to *simulate* M=500 behavioral observations and derive (α, β) from them as pseudo-observations, which is more natural for behavioral proportions. (iii) *Primary metric.* AutoElicit measures predictive accuracy at fixed n; our primary metric is the *sample size reduction ratio* $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}}$—a quantity not reported by any prior work. (iv) *ESS-calibrated prior weight.* AutoElicit assigns no principled weight to the LLM prior; we scale pseudo-observation count by w calibrated to the equivalent sample size of Gao et al. [2026], grounding prior strength in a measurable quantity. AutoElicit also does not adopt SVGD, the no-leakage framework of Ludwig et al. [2025], or a systematic prior-misspecification analysis.

Gouk and Gao [2024] (described in Capstick et al. 2025) use LLMs to generate synthetic feature–label pairs from public datasets, which implicitly define a prior through a separate likelihood term during training. This is *not* a Bayesian prior in the usual sense: LLM-generated data enters the likelihood, not the prior, so there is no prior/likelihood role separation. Capstick et al. note this makes the method susceptible to LLM data memorization on public benchmarks; our held-out Horton data and explicit role separation avoid this failure mode.

Huang [2025] propose LLM-BI, a proof-of-concept pipeline in which an LLM translates a *user's natural-language belief description* into prior distributions and a likelihood function for Bayesian linear regression. LLM-BI validates that posteriors from LLM-specified priors agree with posteriors from manually specified priors on synthetic data. Our work differs on three points. (i) LLM-BI relies on the *user* to supply prior beliefs in natural language; we rely on the *LLM's own behavioral knowledge* from pretraining, without any user-provided prior information. (ii) LLM-BI is evaluated on synthetic data with no human experimental data; our evaluation uses real dictator- and ultimatum-game data. (iii) LLM-BI does not address sample-size reduction, prior calibration, training-data leakage, or failure-mode robustness.

2 Description of the Proposed Project

2.1 Problem Statement and Significance

We consider Bayesian inference over human behavioral parameters in economic experiments, with an LLM serving as an informative prior. Let $\theta \in \Theta$ parametrize the distribution of human behavior—in the dictator game, $\theta = (\alpha, \beta)$ are the shape parameters of a Beta likelihood $p(y \mid \theta) = \text{Beta}(y; \alpha, \beta)$ for sharing ratios $y \in [0, 1]$. Given n human observations $\mathbf{y} = \{y_1, \ldots, y_n\}$, the posterior is:

$$p(\theta \mid \mathbf{y}) \;\propto\; p(\mathbf{y} \mid \theta) \cdot p(\theta). \tag{1}$$

Our quantity of interest (QoI) is the posterior $p(\theta \mid \mathbf{y})$ and, operationally, the minimum sample size n^* required to reach a target posterior precision σ^2_{target}. The central UQ question is: *how much does an LLM-informed prior reduce n^* relative to a flat prior?*

Human experiments are costly (\$10–50 per participant on MTurk), yet Gao et al. [2026] show LLM knowledge is worth dozens to hundreds of human observations for certain economic variables. Our framework assigns LLM predictions and human data to *strictly separate* roles in (1): LLM outputs enter only through the prior, human data only through the likelihood. This avoids the circularity of calibrating an LLM against human data and then using it to predict human behavior—the role-separation also directly implements the no-leakage requirement of Ludwig et al. [2025]. One caveat: a modern LLM trained after January 2023 may have seen Horton's published results; we treat this as a failure mode and address it in the sensitivity analysis.

2.2 Methodology

Prior construction. For each experimental condition x, we query GPT-4o (`gpt-4o-2024-08-06`) with a natural-language description of the game and collect $M = 500$ simulated sharing ratios $\{\tilde{y}_1, \ldots, \tilde{y}_M\}$. We treat these as M pseudo-observations and use them as sufficient statistics to initialize a conjugate Beta prior over $\theta = (\alpha, \beta)$, scaled by an effective weight $w \in (0, 1]$ calibrated to the equivalent sample size of Gao et al. [2026]. This yields the LLM-informed prior $p_{\text{LLM}}(\theta \mid x)$.

Baseline. A flat prior $p(\theta) \propto 1$ combined with MCMC (Metropolis–Hastings) provides the posterior $p(\theta \mid \mathbf{y})$. We estimate the minimum sample size n^*_{flat} by running inference at $n \in \{5, 10, 20, 50, 100, 200\}$ and interpolating the threshold where posterior variance on $\mu = \alpha/(\alpha + \beta)$ drops below $\sigma^2_{\text{target}} = 0.1 \times \text{Var}_{\text{prior}}(\mu)$.

Frontier method. We substitute $p_{\text{LLM}}(\theta)$ for the flat prior and use Stein Variational Gradient Descent (SVGD) for posterior inference. Although the present problem is low-dimensional ($\theta \in \mathbb{R}^2$), SVGD is adopted as the frontier method because it generalizes naturally to multiple experimental conditions and higher-dimensional behavioral parameter spaces; the particle-based representation also makes posterior geometry immediately visualizable. We obtain n^*_{LLM} by the same interpolation procedure.

Evaluation. Our primary metric is the **sample size reduction ratio** $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}}$, where $\rho < 1$ indicates cost reduction. We additionally evaluate:

- *Posterior calibration*: empirical coverage of 95% credible intervals on a held-out split of the Horton data.

- *Compute–accuracy tradeoff*: wall-clock time and ESS for SVGD vs. MCMC at matched accuracy.
- *Failure mode*: we artificially shift the LLM prior mean by $\pm 1, 2$ standard deviations and measure coverage degradation, identifying conditions under which the informative prior is harmful.

The framework is applied to the dictator game and ultimatum game from Horton [2023].

Compute plan and risks. LLM queries require $M \times K \approx 500 \times 4$ API calls (two games, two conditions each), costing under \$5. MCMC and SVGD runs are CPU-only and complete in minutes for a 2-parameter model. The primary risk is LLM prior misspecification (e.g., due to training-data leakage from Horton's published results); this is explicitly studied in the failure-mode evaluation above.

2.3 Contributions

We contribute a Bayesian framework that incorporates LLM simulations as informative priors with strict prior/likelihood separation, an empirical characterization of sample-size reduction ρ across multiple game settings, and a robustness analysis identifying when LLM priors help versus hurt posterior inference.

2.4 Plan and Timeline

Date	Milestone	Deliverable
Mar 2	Proposal submission	This document + 3–4 slides
Mar 11	Check-in 1	Baseline implemented (flat prior + MCMC) on Horton (2023) dictator game; initial posterior results
Apr 8	Check-in 2	LLM prior construction + SVGD implemented; preliminary comparison to baseline
Apr 27	Final presentation	15-min talk with full evaluation results
Finals	Final report	10-page report + reproducible code

Table 1: Project timeline.

References

Alexander Capstick, Rahul G. Krishnan, and Payam Barnaghi. Autoelicit: Using large language models for expert prior elicitation in predictive modelling. In *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267, 2025. arXiv:2411.17284.

Wayne Gao, Sukjin Han, and Annie Liang. How well do llms predict human behavior? a measure of their pretrained knowledge. January 2026. University of Pennsylvania, University of Bristol, Northwestern University.

Henry Gouk and Boyan Gao. Automated prior elicitation from large language models for Bayesian logistic regression. In *The 3rd International Conference on Automated Machine Learning (AutoML 2024)*, 2024.

George Gui and Olivier Toubia. The challenge of using llms to simulate human behavior: A causal inference perspective. November 2025.

John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? *arXiv preprint arXiv:2301.07543*, January 2023.

Yongchao Huang. LLM-BI: Towards fully automated Bayesian inference with large language models. *arXiv preprint arXiv:2508.08300*, 2025.

Jens Ludwig, Sendhil Mullainathan, and Ashesh Rambachan. Large language models: An applied econometric framework. Technical Report 33344, National Bureau of Economic Research, January 2025. URL http://www.nber.org/papers/w33344. Revised December 2025.

Ola Shorinwa, Zhiting Mei, Justin Lidard, Allen Z. Ren, and Anirudha Majumdar. A survey on uncertainty quantification of large language models: Taxonomy, open research challenges, and future directions. *ACM Computing Surveys*, 58(3):63, September 2025. doi: 10.1145/3744238.